

Contents

1 week 3: python 3: loops, logic, errors	1
1.1 review	1
1.2 review challenge	1
1.3 loops	2
1.4 functions	2
1.5 BREAK	3
1.6 logic	3
1.7 csv module	3
1.7.1 Combining logic, loops, and csv:	4
1.7.2 Challenge (that I'll walk you through)	4
1.8 next week: web scraping!	5

1 week 3: python 3: loops, logic, errors

Agenda

- review last week's lessons on lists
- lessons for today: loops, logic
- start working with CSV module using VS Code & Copilot

1.1 review

- lists
 - list indexing vs slicing
- methods vs functions
 - what are some functions and methods we learned?
 - functions are independant
 - methods depend on objects (also called properties, attributes)
 - can tell in syntax

1.2 review challenge

Create a list of words from a text, and do things to the list using list methods.

1.3 loops

How we do things to data.

- types of data for categorizing data; variables for saving data; how to work with lists of data; now, how to do things to lists/groupings.
 - also works with strings
- syntax: for item in collection: print(item).
 - practice with both lists and strings
- a note on variable names:
 - the variable following "for" is assigned on the fly
- f-strings
- CHALLENGE: prime numbers

Doing things to strings

- how to do things to strings within loops
- 'HELLO'.lower()
- make a list of cities, and make them all lowercase
- now save that list to a new list, an empty list
 - why would we want to do this?

1.4 functions

- functions do things to data
 - functions in algebra $f(x) = x + 1$
- two steps of a function: defining, calling
 - how to define a function to add a number
- calling the function
- challenge: how to make a function that prints a string with your name?

1.5 BREAK

1.6 logic

Boolean data

- `type()` -> True or False
- evaluates mathematical expressions
 - different operators, look them up. Many different kinds.
- if statement for checking age
 - multiple conditions

Combining loops with logic:

- DEFENDING WOMEN FROM GENDER IDEOLOGY EXTREMISM AND RESTORING BIOLOGICAL TRUTH TO THE FEDERAL GOVERNMENT
- if it contains the word "gender", "protect", or others, we will print.

```
sentences = text.split('.')
for i in sentences:
    if 'binary' in i:
        print(i)
```

1.7 csv module

- csv module
 - what is a module? a collection of code for doing something, in this case, for opening csv files
 - read a little of the docs on CSV module, reader
- printing rows from csv on Campaign Violations:
 - `import csv`
 - open the file with open statement
 - print the rows

```

with open('./Downloads/Enforcement_Actions_Board_Determinations_and_Penalties_20250210
    data = csv.reader(f)
    for row in data: # can we slice this to do only the first 5 rows?
        print(row)

```

How do we get just the first object from each column? The names?

- breakup the problem into parts
- check the type of data(s)
- how do we access info from a list?

```

with open ('violations_sample.csv', 'r') as f:
    data = csv.reader(f)
    for row in data:
        print(row[1])

```

1.7.1 Combining logic, loops, and csv:

- search for a specific candidate name in the dataset
- print out all rows containing that candidate's name
- advanced: print out only the date and the violation

```

with open('violations.csv', 'r') as f:
    data = csv.reader(f)
    for row in data:
        if "Eric" in row[1]:
            print(row)

```

1.7.2 Challenge (that I'll walk you through)

Combine what we know from the above with f-strings to write more complex output.

Write a loop that prints out the Candidate's name and Violation if that violation contains the word "contribution" in it. Use f-strings so that you can format the answer the following:

Name: [candidate name], Violation: [candidate violation]

Here's the answer, but don't look until you've spent at least 5 minutes working on it!

```
with open ('violations.csv', 'r') as f:
    dict_reader = csv.reader(f)
    for row in dict_reader:
        violation = row[3]
        if "contribution" in violation:
            candidate = row[1]
            print(f'Name: {candidate}, Violation: {violation}')
```

1.8 next week: web scraping!